

Alarm Indication Not Received When Alarm Activated

Symptoms

- The engine room panel or remote panel does **not** indicate an alarm when active.

How To Use This Tree

This symptom tree can be used to troubleshoot engine alarm symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Logic Unit	
	STEP 1A. Check Customer Interface Box Logic Unit Red Lamp Indicator	Is red lamp indicator illuminated?
STEP 2.	Check Panel Wiring	
	STEP 2A. Check Engine Room Panel Wiring	Is alarm lamp illuminated?
	STEP 2A-1. Check Engine Room Panel Alarm (Red Lamp) Supply Wire	Less than 10 ohms resistance?
	STEP 2A-2. Check the Engine Room Panel Return Wire	Less than 10 ohms resistance?
STEP 3.	Check Engine Room Panel Cable	
	STEP 3A. Check Engine Room Panel Cable for Open Circuit	Less than 10 ohms resistance?
STEP 4.	Check Customer Interface Box Wiring Associated with Engine Room Panel	

STEPS		SPECIFICATIONS
	STEP 4A. Check Engine Room Panel Alarm (Red Lamp) Supply Wire	Less than 10 ohms resistance?
	STEP 4B. Check the Engine Room Panel Return Wire	Less than 10 ohms resistance?
STEP 5.	Check Remote Panel	
	STEP 5A. Check Remote Panel Alarm Lamp	Is alarm lamp illuminated?
STEP 6.	Check Remote Panel Cable	
	STEP 6A. Check Remote Panel Cable for Open Circuit	Less than 10 ohms resistance?
STEP 7.	Check Customer Interface Box Wiring Associated with Remote Panel	
	STEP 7A. Check the Remote Panel Alarm (Red Lamp) Supply Wire	Less than 10 ohms resistance?
	STEP 7B. Check the Remote Panel Return Wire	Less than 10 ohms resistance?
STEP 8.	Check Customer Interface Box Wiring Associated with Red Lamp Driver	
	STEP 8A. Check Red Lamp Signal Wire	Less than 10 ohms resistance?
STEP 9.	Check Customer Interface Box to Engine Harness Cable	
	STEP 9A. Check Red Lamp Signal Wire	Less than 10 ohms resistance?
STEP 10.	Check For Failed Component	
	STEP 10A. Check Customer Interface Box Logic Unit and Control Panel	Red lamp illuminated?

STEP 1. Check Customer Interface Box Logic Unit

STEP 1A. Check Customer Interface Box Logic Unit Red Lamp Indicator

Conditions :

- Locate customer interface box
- Open customer interface box.

Action	Specification/Repair	Next Step
Verify red lamp indicator is illuminated.	Is red lamp indicator illuminated? YES	2A
	Is red lamp indicator illuminated? NO	8A

STEP 2. Check Panel Wiring

STEP 2A. Check Engine Room Panel Wiring

Conditions :

- Locate engine room panel

Action	Specification/Repair	Next Step
Verify alarm red lamp is illuminatated.	Is alarm lamp illuminated? YES	5A
	Is alarm lamp illuminated? NO	2A-1

STEP 2A-1. Check Engine Room Panel Alarm (Red Lamp) Supply Wire

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
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Check wires between harness connector and control panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel alarm (red lamp) supply pin on connector C14. • Place the other test lead on the engine room panel alarm (red lamp) supply pin on the control panel connector.	Less than 10 ohms resistance? YES	2A-2
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 2A-2. Check the Engine Room Panel Return Wire

Conditions : • Locate engine room panel • Open engine room panel door.		
Action	Specification/Repair	Next Step
Check wires between harness connector and control panel connector. • Disconnect cable C14 from the engine room panel. • Connect one test lead on the engine room panel return pin on the on connector C14. • Place the other test lead on the engine room panel return terminal on the control panel connector.	Less than 10 ohms resistance? YES	3A
	Less than 10 ohms resistance? NO Repair : Replace the fault wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 3. Check Engine Room Panel Cable

STEP 3A. Check Engine Room Panel Cable for Open Circuit

Conditions :

- Disconnect cable connector C14 from the engine room panel
- Disconnect cable connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
Check the engine room panel cable. • Install a jumper between engine room panel alarm (red lamp) supply pin and the engine room panel return supply pin in connector C14. • Place one test lead in the engine room panel alarm (red lamp) supply pin in connector C7. • Place the other test lead in the engine room panel return pin in connector C7.	Less than 10 ohms resistance? YES	4A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 4. Check Customer Interface Box Wiring Associated with Engine Room Panel

STEP 4A. Check Engine Room Panel Alarm (Red Lamp) Supply Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
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Check the engine room panel alarm (red lamp) supply wire. - Place one test lead on the engine room panel alarm (red lamp) supply pin in connector C7. - Place the other test lead on the engine room panel alarm (red lamp) supply terminal on the customer interface box logic unit.	Less than 10 ohms resistance? YES	4B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 4B. Check the Engine Room Panel Return Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
Check the engine room panel return wire. - Place one test lead on the engine room panel return pin in connector C7. - Place the other test lead on the battery voltage return pin on the customer interface box logic unit.	Less than 10 ohms resistance? YES	5A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 5. Check Remote Panel

STEP 5A. Check Remote Panel Alarm Lamp

Conditions :

- Locate remote panel.

Action	Specification/Repair	Next Step
Verify red alarm lamp is illuminated.	Is alarm lamp illuminated? YES	10A
	Is alarm lamp illuminated? NO	6A

STEP 6. Check Remote Panel Cable

STEP 6A. Check Remote Panel Cable for Open Circuit

Conditions :

- Locate and open customer interface box
- Disconnect remote panel cable from customer interface box X4 connector
- Locate and open remote panel.

Action	Specification/Repair	Next Step
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Check the remote panel cable. • Install a jumper between remote panel alarm (red lamp) supply terminal and the remote panel return terminal on remote control panel X4 in the remote control panel. • Place one test lead on the remote panel alarm (red lamp) supply terminal in the remote panel cable. • Place the other test lead on the remote panel return terminal in the remote panel cable.	Less than 10 ohms resistance? YES	7A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 7. Check Customer Interface Box Wiring Associated with Remote Panel

STEP 7A. Check the Remote Panel Alarm (Red Lamp) Supply Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step

Check the remote panel alarm (red lamp) supply wire. - Place one test lead on the remote panel alarm (red lamp) supply terminal of terminal strip X4. - Place the other test lead on the remote panel alarm (red lamp) supply terminal in the customer interface box logic unit.	Less than 10 ohms resistance? YES	7B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 7B. Check the Remote Panel Return Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
Check the remote panel return wire. - Place one test lead on the remote panel return terminal of terminal strip X4. - Place the other test lead on the remote panel return terminal on the customer interface box logic unit.	Less than 10 ohms resistance? YES	10A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 8. Check Customer Interface Box Wiring Associated with Red Lamp Driver

STEP 8A. Check Red Lamp Signal Wire

Conditions :

- Disconnect cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check red lamp signal wire. • Place one test lead on the red lamp signal pin at the C3 connector. • Place the other test lead in the red lamp signal terminal of the customer interface box logic unit.	Less than 10 ohms resistance? YES	9A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 9. Check Customer Interface Box to Engine Harness Cable

STEP 9A. Check Red Lamp Signal Wire

Conditions :

- Disconnect cable connector C1 from the customer interface box
- Disconnect cable connector C8 from the engine harness.

Action	Specification/Repair	Next Step

Check red lamp signal wire. • Install a jumper between red lamp signal pin and coolant level 1 (keel cooled) return (sensor return 1) pin in the C8 connector. • Place one test lead in the red lamp signal pin of the C1 connector. • Place the other test lead in the coolant level 1 (keel cooled) return (sensor return 1) pin of the C1 connector.	Less than 10 ohms resistance? YES	10A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 10. Check For Failed Component

STEP 10A. Check Customer Interface Box Logic Unit and Control Panel

Conditions :

- Open customer interface box
- Make sure all panel system cables are connected.

Action	Specification/Repair	Next Step

Check red lamp indicator on customer interface box logic unit.	Red lamp illuminated? YES Repair : Replace control panel. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html) or 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.
	Red lamp illuminated? NO Repair : Replace the customer interface box logic unit after verifying on-engine harness and engine control module are operating properly. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.